

WHAT'S NEW

Ancient frog species discovered in Kerala

The Western Ghats in India are a biodiversity hotspot, with hundreds of species that are found nowhere else in the world. A team of researchers led by SP Vijayakumar from George Washington University and Kartik Shanker from the Indian Institute of Science, have identified a new species of frog which is the only known specimen of its genus and subfamily. The species is called *Astrobatrachus kurichiyana*, or the Starry Dwarf Frog, and was found in one of the nocturnal expeditions of the team in the Wayanad Plateau. The frog did not get much attention for a few years, during which time a research team was assembled to describe the



new species. The openVertebrate project allows CT scans of skeletons to be shared with researchers and even the common

public, which was helpful in the analysis of the new frog. Analysis of the CT scans and the DNA of the frog showed that it belonged to an entirely new species, and diverged from its closest relatives more than ten million years ago. The frog helps scientists better understand the Amphibian tree of life.

<https://dgit.in/swf>

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Space is all over our science section this month, with Space Roar and the technicalities of a Mars landing being explored. We also analyse the possibility of an HIV cure as well as the connection between your brain and your gut!



Children are sometimes smarter than adults

As young kids are more preoccupied with exploring and learning about the world, they are better at solving some problems than adults, who are more focused on accomplishing particular goals. Researchers are collaborating with engineers to make AI smarter by incorporating curiosity. <https://dgit.in/smart-babby>

Researchers recreate alien atmospheres

Scientists at NASA's JPL have recreated the atmospheres of a particular category of exoplanets, within a lab. Hot Jupiters are gas giants in close orbit around their host stars. Their size means that they are quite easily discovered. The temperatures on these can rise beyond 2,800 degrees Celsius, and the researchers created these extreme temperatures using an oven. A mixture of hydrogen gas and

0.3 percent carbon monoxide gas, which is expected to be found considering current



theories of planetary formation, was heated. The key element was exposure to UV rays, which can be expected

from the host star. The UV light led to the formation of aerosols. The finding matches with observations, where the higher altitudes of Hot Jupiters showed opaque banks of clouds, which scientists believed were difficult to form at the low pressure regions of the higher altitudes. Scientists now understand how these hazy atmospheres were formed.

<https://dgit.in/alienskies>



Geoengineering dosage

Scientists are seriously considering geoengineering to alleviate global warming. One way is to dump aerosols in the air to reflect sunlight. It could disrupt rainfall around the world, but is viable if smaller doses of aerosols are used, along with reduced emissions. <https://dgit.in/cloudseas>



Eyes indicate mistakes

Researchers rigged a test where participants had to decide which ear they were hearing a click on. It was discovered that the pupil size changed when they were making a wrong decision. The eyes can indicate when we make suboptimal decisions. <https://dgit.in/eyefail>



Tiny tyrannosaur discovered

The fossil record of the tyrannosaur's ascent to the top of the food chain has been sparse. A relatively tiny tyrannosaur ancestor has been dated to about 96 million years ago, which shows they took only about 15 million years to become apex predators. <https://dgit.in/tinytrex>